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Chatbots on a Mid-scale Knowledge Base: Using an R2 University as an Example

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***Abstract*— While LLMs like ChatGPT, Claude, and Gemini have proven effective at handling common, everyday tasks, they often lack the domain-specific expertise necessary for more specialized consultations. This limitation makes it essential to integrate a targeted knowledge base when developing chatbots for specific domains. In this study, we investigated the performance of chatbots utilizing large language models (LLMs) within the context of a mid-scale knowledge base for a regional R2 university in the U.S. Using a web-crawled knowledge base, we created chatbots across multiple platforms, testing them against a set of predefined questions to evaluate correctness and comprehensiveness. Our findings indicate that while ChatGPT models, particularly those augmented with both a knowledge base and real-time search capabilities, provide the most balanced responses on objective questions, hallucinations remain an issue. In the subjective testing questions, no single chatbot emerged as the best performer, with different models excelling in various areas based on the personas and questions. The study suggests that while LLMs are valuable tools in education, careful consideration of their limitations is necessary for improving their application in domain-specific consulting.**

Keywords-chatbot; large language models (LLMs); knowledge base; LLM evaluation framework

I. INTRODUCTION

With the rapid development of artificial intelligence (AI), Large Language Models (LLMs), as a major breakthrough in the field of AI, are gradually changing the way people acquire, understand, and use information. These models, such as ChatGPT [1], Claude [2], Gemini [3] etc., greatly enhance the efficiency and convenience of users obtaining information through the internet due to their powerful natural language processing and knowledge generation capabilities. In the field that requires domain knowledge and understanding of the context/scenario, such as consultation, LLMs have demonstrated "remarkable potential" [4]. In education settings, LLMs are also "promising tools for open education" as they can provide customized and interactive assistance to students and thereby improve the independence and autonomy of the learners [1]. These types of "domain-specific" LLMs [5] can not only quickly respond to the diverse needs of students, but also provide personalized advice and information based on massive data, greatly reducing the burden of traditional consultation/supporting/feedback methods.

However, despite the enormous potential demonstrated by LLMs, their reliability and accuracy issues in practical applications are becoming increasingly prominent. Due to the complexity, diversity, and potential bias of model training data, LLMs may experience misunderstandings, erroneous reasoning, and even fabricated information during the understanding and inference process [6] (sometimes referred to as "hallucination"). This uncertainty undoubtedly poses potential risks for users who rely on model recommendations to make important decisions. Therefore, an in-depth exploration of the accuracy and reliability of LLMs in providing consulting information is of great significance for promoting healthy and safe applications in different domains.

This study aims to systematically evaluate the performance of different major LLMs in answering a common set of questions. Specifically, we utilized web crawling technology to extensively collect information on University of North Carolina Wilmington (UNCW), a regional R2 university on the east coast of the United States as our knowledge base. The evaluation is done based on both objective questions (about new student enrollment, such as tuition fees, accommodation, and food) and subjective questions (using given scenarios/personas). Subsequently, we input this data into multiple commercialized large language modeling platforms (ChatGPT, Claude, Gemini, and Copilot) to create Chatbots that can simulate answering questions in real-life scenarios based on the given knowledge base.

By comparing and analyzing the answers provided by different LLMs, this study aims to achieve the following objectives: 1) demonstrate our framework of LLM evaluations using a knowledge base and a set of questions, both objective and subjective; 2) explore the impact of a reliable knowledge base versus/with online search in building a domain-specific chatbot, and analyze the role and limitations of them in improving the quality of answers; 3) based on the research results, strategies and suggestions for improving the application of LLMs in the field of educational consulting are proposed,

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providing reference for practitioners and researchers in related fields.

II. Related work

Chatbots are an important research method in the field of natural language processing (NLP). In the early field of machine translation which is a subdomain of NLP, BLEU [7] and METEOR [8] were two common evaluation metrics. BLEU evaluates the accuracy of the translated text by calculating the similarity between the machine-generated translation and a set of reference translations. METEOR improves the BLEU algorithm by introducing an alignment algorithm to evaluate the similarity between the generated translation and the reference translation, which can better handle synonyms and word order differences. However, these evaluation metrics are ineffective for evaluating chatbots. In 2020, Mehri and Eskenazi proposed the FED measurement metric [9], which can measure fine-grained dialogue quality at both the dialogue turn (defined as "a dialog context and a system response (from chatbot)" [9]) level and the entire dialogue level. FED was reported to achieve a moderate to strong correlation with human judgments at both levels. However, with the advancement of chatbot technology, especially after the introduction of large language models (LLMs), people have put forward higher requirements for chatbots. People's focus on dialogue has shifted from fluency to multi-dimensional evaluations of chatbots' reading comprehension, reasoning capability, math, and other technical measures [10].

Researchers utilized specific datasets to evaluate the fluency, naturalness, and other aspects of dialogue response, testing whether the dialogue level of chatbots can reach human levels. For example, Mendonça et. al introduced a new evaluation benchmark dataset SODA-EVAL [11], it used more than 120000 turn-level assessments for training, covering 10000 conversations. Human validation and annotation tasks were also conducted to confirm the quality of automatic annotation. A rating system ranging from 1 to 5 was used to evaluate the quality of dialogue responses. The evaluation focuses on the effectiveness of dialogue annotations, including whether they contradict the information in the dialogue history, whether the model can cover all relevant information in the conversation, and the fluency and naturalness of dialogue response, which includes evaluating the common sense, participation, and repetition in the conversation. This research found that although GPT-4 performs well in many aspects, there is still room for improvement in coherence and common-sense reasoning. This indicates that although large language models have made significant progress in generating fluent and relevant responses, they still face challenges in simulating the complexity of human dialogue.

Researchers have reported successes in creating domain-specific chatbots that can support domain experts in reading and making decisions. The progress of domain-specific chatbots also entices research in domain-specific chatbot evaluations. In the work by Song et al., the authors developed a questionnaire of 21 questions and two clinical scenarios related to urolithiasis [4]. They tested four LLMs (Claude, Bard, ChatGPT4, and Bing). Their domain experts then evaluated the model responses regarding accuracy, comprehensiveness, legibility, human care, and clinical case analysis ability based on a 5-point Likert scale. They reported that Claude and GPT-4 were the top-performing LLMs in their evaluation matrices. However, they only evaluated urolithiasis-related dialogs, which is a relatively small domain.

In the field of education, Hwang et. al [12] used an AI-driven approach to create and assess multiple-choice questions in the fields of chemistry and biology. They evaluated the quality of the questions by using the Item Writing Flaw (IWF) criteria [13] and combined machine learning models with human assessments to verify the alignment of the questions with Bloom's Taxonomy. The study utilized the RoBERTa model, a machine learning model, to validate 120 generated questions. A domain expert with over 28 years of experience in STEM education assessed 57 of those questions. The research indicated that GPT-3.5 can generate questions aligned with Bloom's Taxonomy levels. However, there are differences in the quality assessment of the questions between human and machine evaluations. This suggests that there is still a discrepancy between machine learning models and human assessments when evaluating AI-generated content. Thus, a carefully considered set of standards is necessary for evaluating the quality of AI-generated materials.

III. Project Scope and Data

The scope of our work involves building chatbots on different platforms using data related to UNCW, an R2 university on the East Coast of the U.S. The data is crawled mainly from URLs in the format "https://uncw.edu/***." We also compare the performance of the chatbots when they answer evaluation questions with or without restricted Internet access. In the data acquisition process, a web crawler was used to gather content from multiple pages within the domain. The web crawling process ensures that the pages cover comprehensive information about this university, including details on housing, transportation, safety reports, meal plans, tuition fees, scholarships, and student life. The crawler organized the content into a total of 20 topic-specific files, which are in the format of .docx, .pdf, or .csv, generating more than 148,000 tokens, which served as the foundation for the chatbot's knowledge base. (By comparing two chatbots' performance using the same knowledge base, but one as a .pdf and another as a .docx file, we noticed that the chatbots who received the .docx as knowledge base performed better. Therefore, in the rest of this paper, the chatbots were trained with .docx files as a knowledge base unless noted separately.)

In addition to data collection, various commercialized AI platforms and chatbot frameworks were explored to identify the most suitable tool to test/compare for this project. GPT-4o, Gemini 1.5, Claude, and Copilot Studio were selected for this project. We created multiple chatbots and provided our knowledge base to them for testing. A chatbot, GPT-4o, without the knowledge base, is also tested as a baseline. We also analyzed the impact of allowing the chatbot to access the Internet for searches based on the knowledge base.

When selecting the commercialized LLMs, we focused only on the independent models/model providers. Therefore, the "secondary" AI platforms that use those LLMs (such as BoodleBox) were excluded. We also attempted to include Meta AI, but building a chatbot with an additional knowledge base

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from their web user interface has not yet been supported. Table I lists the chatbots we have tested or selected.

TABLE I. LLMS STUDIED IN OUR PROJECT

Chatbot	Domain-based chatbot function with knowledge base support	Cost	Search online in real-time?	Additional constraints
ChatGPT	My GPTs	$20/month	yes	Up to 20 files as the knowledge base
Gemini	Vertex AI	Free (Gemini 1.5 flash)	no	All the input knowledge base should be in one category - 1) PDF or doc, 2) JSON, and 3) CSV; 10 files, 1 million line window size
Claude	Claude Pro projects	$20/month	no	Input knowledge base total line limit (~20,000-40,000) because of Claude's maximum context window size
Copilot	Copilot studio	$200/month	Yes (but only from 4 links)	Can only support four URLs, two Dataverses (up to 15 tables), supports file uploads
Meta AI (Llama)	Need to set up your own environment	free	no	Meta AI web interface does not support knowledge base uploads

When testing these LLMs, to minimize the impact of irrelevant factors, we provided the same prompt for the chatbots on each platform and conducted the tests using identical questions.

IV. EVALUATION FRAMEWORK AND RESULTS

We developed a structured framework to assess chatbot performance with two test sets. The first set, comprising 20 objective questions, was used in single-turn interactions (where the user asks a single question and receives a single response without any follow-up or context retention) to evaluate the performances of the chatbots. The second set introduced three fictional personas; for each, a chatbot will answer 13 subjective questions through multi-turn dialogues (where multiple exchanges occur between the user and the chatbot, allowing for context retention and follow-up questions within a coherent conversation).

A. Evaluation framework

We created two test sets, both covering various aspects of the university. The first set of 20 objective questions focused on factual aspects such as tuition/fees, housing, and campus safety. The second set included three fictional personas and 13 subjective questions that were related to personal preferences, such as "What food options align with my taste does the on-campus dining offer?" and "Are there any on-campus housing options that will fit my lifestyle and preferences?" To create these personas, we developed profiles of three virtual incoming college students, detailing their preferred majors, food preferences, personal hobbies, and other individualized characteristics.

When using the first set of questions to test the chatbot, we conducted single-turn conversations to obtain and evaluate responses directly. In contrast, for the second set, we employed a multi-turn dialogue approach. First, we provided the fictional persona's background information to the chatbot to provide context for the interaction. We then asked subjective questions from the test set, expecting the chatbot to respond based on both the provided knowledge base and the specific details of the persona. This approach allowed us to assess the chatbot's ability to handle both straightforward questions and more complex, context-driven conversations by integrating its knowledge with personalized responses, offering a comprehensive view of its adaptability and effectiveness.

When we evaluated the dialogue with the chatbots, we focused on two aspects, *correctness* and *comprehensiveness*. For each response from the chatbots, we rate the correctness and comprehensiveness using a 3-point Likert scale. The Likert scales were defined as follows:

Correctness -

- 0 (Poor) - wrong information (hallucination) or refusal to answer when the knowledge base has included this information.
- 1 (Fair) - partially wrong/inaccurate or misleading information provided or refuse to answer when the knowledge base does not provide adequate information to answer.
- 2 (Good) - accurate information, or the information is very close to the information provided (considering some numbers, like student population, will change every year).

Comprehensiveness -

- 0 (Poor) - not enough information is provided to cover all the aspects of the question.
- 1 (Fair) - some information related to the question but inadequate, or too much additional and unrelated information.
- 2 (Good) - good coverage of all the aspects, no or limited unrelated information provided.

The last two authors of this paper rated all the responses written by our chatbots. Both of them have been teaching, studying, or working at UNCW for at least three years.

B. Experimental results

Objective question:

We created three different ChatGPT-based chatbots: ChatGPT4o without knowledge base (**GPT-4o**), ChatGPT4o with knowledgebase but no online search (**GPT-4o+KB**), and ChatGPT4o with knowledge base and search (**GPT4o+KB+S**). GPT-4o is the baseline model in our test, and without the knowledge base, it used its built-in knowledge and real-time searches in our tests. We found that **GPT-4o** was able to produce reasonably accurate and comprehensive answers that are comparable to **GPT-4o+KB** and better than Gemini and Copilot

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Studio. However, **GPT-4o** may make mistakes because the searched information may not be from the university website/domain. The ChatGPT-based model that uses the given knowledge base but no real-time searches, **GPT-4o+KB**, could produce a similar level of correctness, but we observed that its performance of comprehensiveness was slightly weaker. The ChatGPT-based model that uses both the given knowledge base and online searches (the prompt required it to use the knowledge base as the primary information source), **GPT4o+KB+S**, produced the highest correctness and comprehensiveness among all the models.

Claude provided the most accurate result. Considering that Claude is an offline LLM that does not support real-time online searches, we found that Claude's correctness was phenomenal (the tied highest). However, we also found that Claude tended to provide additional information that was sometimes unrelated to the question. Therefore, Claude's comprehensiveness score was the second highest, mainly due to unrelated information. However, it's also worth mentioning that Claude supports a relatively small knowledge base, and as a result, we had to separate our knowledge base files and produce four chatbots on Claude before they could provide answers to all test questions.

We found that our chatbot based on **Gemini** missed important information that was actually provided in the knowledge base. It may also not provide a comprehensive answer from all the possible perspectives. For example, when we tested it using a question about the tuition, it only provided answers for in-state tuition but missed the out-of-state tuition. In addition, Gemini did not take tabular data (such as Excel files or CSV files), therefore part of our knowledge base (major list and the core course list for each major) could not be taken by Gemini, leaving several "n/as" on the experiment results.

The chatbot based on **Copilot Studio** received the greatest number of 0s, signifying the lack of capability to provide the answer even if it was included in the knowledge base. Compared to other chatbots, Copilot also provides shorter and less comprehensive responses. Considering that Copilot Studio allows searches on up to four hyperlink domains, we expected it to perform better when using both the given knowledge base and real-time searches, but it was the weakest model in terms of both correctness and comprehensiveness. However, Copilot Studio had the most extensive features, such as customizing conversational flow for different topics. It still has the potential to handle more complex tasks in the future.

Table II lists the scores we have given to our chatbot on correctness and comprehensiveness.

TABLE II. PERFORMANCE OF OUR CHATBOTS ON OBJECTIVE QUESTIONS

	correctness						***comprehensiveness***					
Question	***GPT-4o***	***GPT4o+ KB***	***GPT4o+ KB+S***	***Claude***	***Gemini-1.5-flash***	***Copilot Studio***	***GPT-4o***	***GPT4o+ KB***	***GPT4o+ KB+S***	***Claude***	***Gemini-1.5-flash***	***Copilot Studio***
Q1	2	2	2	2	2	2	2	2	2	1	1	2
Q2	2	2	2	2	2	0	1	1	2	1	2	1
Q3	2	2	2	2	2	2	2	2	1	1	2	2
Q4	1	1	2	2	1	1	0	0	2	2	1	1
Q5	2	2	2	2	n/a	2	2	2	2	1	n/a	2
Q6	2	0	2	2	2	2	2	1	2	1	2	2
Q7	2	2	2	2	2	2	2	2	2	1	2	2
Q8	1	2	2	2	2	2	2	1	2	2	1	1
Q9	0	1	2	2	1	0	0	1	2	2	1	0
Q10	2	2	1	2	2	0	1	2	1	2	2	0
Q11	2	2	2	2	0	0	2	2	2	1	0	0
Q12	1	1	2	1	0	1	1	1	2	2	1	1
Q13	1	2	1	2	2	2	1	2	2	2	2	2
Q14	2	2	2	2	2	2	2	2	2	2	2	2
Q15	1	2	2	2	2	2	1	2	2	2	2	2
Q16	2	1	2	2	2	2	2	0	2	2	2	2
Q17	2	2	2	2	0	2	2	2	2	2	0	2
Q18	2	2	2	2	2	1	2	2	2	2	1	1
Q19	2	2	2	2	2	1	2	1	2	2	1	1
Q20	2	2	2	2	2	0	2	2	2	1	1	1
Q21	2	1	2	1	0	2	1	1	1	0	0	1
Avg.	1.67	1.67	**1.90**	**1.90**	1.50	1.33	1.52	1.486	**1.86**	1.524	1.30	1.33

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Subjective questions:

We tested the performance of the chatbots on three fictional personas. For the first one, UNCW, the university involved, was not a good fit, considering the intended major, preferred location, and city setting. The second one, by contrast, was clearly a good fit for UNCW, considering the academic plan and personal hobbies specified. The last fictional persona may find UNCW as a "borderline fit" for his academic plan. We created those three functional personas with the purpose of testing the chatbots in answering subjective questions related to each one's specific needs and interests.

Overall, the ChatGPT-based models, **GPT4o+KB** and **GPT4o+KB+S**, had good but sometimes unstable performances on the test of three functional personas. The main source of the 0 ratings comes from hallucinations - they make up information that the knowledge base did not include. For example, **GPT4o+KB+S** suggested the third person come for the "Bachelor of Science in Civil Engineering" degree, which was not on the major list that was part of the knowledge base. We found a similar model that relied on the knowledge base only, **GPT4o+KB**, was less likely to hallucinate.

Claude was the most straightforward one in providing information like "this university is not a good fit" to the first person and the third person, making it the top performer for the first persona. However, we also noticed that sometimes this chatbot was not able to find personal suggestions due to the lack of information (Claude did not support real-time searches online), then, it may provide more general suggestions instead. For example, when answering the question, "Is it easy to rent an off-campus apartment near the school that is aligned to my personal needs?" Claude did not find rich enough information from our knowledge base; therefore, it only provided more general answers about the off-campus housing availability and cost.

Gemini may answer our testing questions while not considering some of the specific person's attributes. For example, when calculating the student loans, the loan estimate was provided based on in-state tuition, while the second person was an out-of-state student. Another concern with comprehensiveness is that it provided long responses to most of the answers that present *facts* but are not necessarily related to this person's *needs*. However, our Gemini-based chatbot performed better than GPT4o and Claude-based models on the third fictional person. In addition, we were not able to use all our testing questions because the tabular data could not be added to Gemini, leaving part of the "n/as" in Tables III-V.

The chatbot based on **Copilot Studio** tended to provide answers that were consistent with the knowledge base in the subjective question testing; however, we found that the answers may not be consistent with the needs of the fictional person. For example, the first persona preferred housing options with "her own apartment/suite with a private bedroom and bathroom." However, Copilot Studio still presented the option of "double-occupancy, pod-style rooms," which was clearly not the option that this person preferred. We also found that the answers written by Copilot Studio were not as comprehensive as other chatbots. When we asked the chatbots to help the fictional persons with estimating the student loan and the payments after graduation, most of the chatbots helped the students with calculations and provided estimated numbers, but Copilot Studio responded – "You can estimate your total student loan payment by multiplying the annual loan amount by the number of years and adding any interest accrued." While this answer is not mistaken, this is not a helpful answer.

TABLE III. PERFORMANCE OF OUR CHATBOTS ON SUBJECTIVE QUESTIONS - THE FIRST PERSONA

	correctness					*comprehensiveness*				
Question	*GPT4o+KB*	*GPT4o+KB+S*	*Claude*	*Gemini-1.5-flash*	*Copilot Studio*	*GPT4o+KB*	*GPT4o+KB+S*	*Claude*	*Gemini-1.5-flash*	*Copilot Studio*
Q1	2	2	2	n/a	0	2	2	1	n/a	0
Q2	2	2	2	2	2	2	2	1	2	2
Q3	2	2	2	2	2	2	2	2	2	2
Q4	2	2	2	2	1	1	1	2	2	1
Q5	2	2	2	2	2	1	2	1	1	1
Q6	0	0	2	2	2	0	0	2	1	1
Q7	0	0	2	n/a	0	1	1	2	n/a	1
Q8	1	1	2	2	1	2	1	2	2	1
Q9	1	2	0	2	1	2	2	2	2	1
Q10	1	2	2	2	2	1	2	2	1	2
Q11	2	2	2	2	1	1	2	1	1	1
Q12	1	1	2	1	1	2	2	2	1	1
Q13	2	2	2	1	1	2	2	2	1	1
Avg.	1.38	1.54	**1.85**	1.82	1.23	1.46	1.62	**1.69**	1.45	1.15

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TABLE IV. PERFORMANCE OF OUR CHATBOTS ON SUBJECTIVE QUESTIONS - THE SECOND PERSONA

	correctness					*comprehensiveness*				
Question	***GPT4o+KB***	***GPT4o+KB+S***	***Claude***	***Gemini-1.5-flash***	***Copilot Studio***	***GPT4o+KB***	***GPT4o+KB+S***	***Claude***	***Gemini-1.5-flash***	***Copilot Studio***
Q2-1	2	2	2	n/a	2	2	2	2	n/a	1
Q2-2	2	2	2	2	2	2	2	2	2	1
Q2-3	2	2	2	2	2	2	2	2	2	2
Q2-4	2	2	2	2	1	1	2	2	1	1
Q2-5	2	2	1	1	2	1	1	2	1	1
Q2-6	1	2	1	1	1	2	2	1	2	2
Q2-7	1	2	1	n/a	n/a	0	2	2	n/a	n/a
Q2-8	1	2	2	1	1	2	2	2	2	0
Q2-9	2	2	2	1	2	2	2	2	2	2
Q2-10	2	2	2	2	n/a	2	1	2	2	n/a
Q2-11	2	2	2	2	0	1	1	1	1	0
Q2-12	1	2	1	1	0	1	2	2	2	1
Q2-13	2	2	2	n/a	1	2	2	1	n/a	1
Avg.	1.69	**2**	1.69	1.50	1.27	1.54	**1.77**	**1.77**	1.67	1.09

TABLE V. PERFORMANCE OF OUR CHATBOTS ON SUBJECTIVE QUESTIONS - THE THIRD PERSONA

	correctness					*comprehensiveness*				
Question	***GPT4o+KB***	***GPT4o+KB+S***	***Claude***	***Gemini-1.5-flash***	***Copilot Studio***	***GPT4o+KB***	***GPT4o+KB+S***	***Claude***	***Gemini-1.5-flash***	***Copilot Studio***
Q3-1	1	1	2	n/a	0	2	2	2	n/a	1
Q3-2	2	1	2	2	1	2	2	2	2	0
Q3-3	2	1	2	2	1	2	1	1	2	0
Q3-4	2	2	2	2	2	2	2	1	2	1
Q3-5	1	1	1	1	2	2	1	1	2	1
Q3-6	0	2	2	1	1	1	2	2	1	1
Q3-7	2	0	1	n/a	0	2	1	2	n/a	0
Q3-8	2	2	1	2	0	2	2	1	1	1
Q3-9	2	2	1	2	1	2	1	1	2	1
Q3-10	1	2	1	2	0	1	1	2	2	1
Q3-11	2	2	2	2	2	2	2	2	2	1
Q3-12	1	1	1	1	1	2	2	2	1	1
Q3-13	2	2	2	2	2	2	2	2	2	2
Avg.	1.54	1.46	1.54	**1.72**	1	**1.85**	1.62	1.62	1.73	0.85

The full assessment questions and results can be found in our GitHub repository. [1]

V. CONCLUSION AND FUTURE WORK

In this paper, we crawled the available information for University of North Carolina Wilmington, a regional R2 university on the east coast of the United States. Considering the different aspects of data on academics, admission, student life, research, university athletics, etc., the volume of the unstructured knowledge base was likely on the same level as a small-to-medium cooperation [14]. In addition, we also included a university annual security report (as a .pdf file), a major list, and a core course list (crawled as an Excel file). We created a testing framework targeting the assessment of the capability of

[1] Github repository -https://github.com/larguncw/ICBAIE_2024_chatbots.

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chatbots with the given knowledge base to answer both objective (factual) and subjective (personal-preference-based) questions from potential incoming students' perspectives.

Among the commercialized LLM models tested in our experiments, we found that ChatGPT4o-based models were the strongest in both correctness and comprehensiveness on objective questions, which is consistent with Rydzewski's findings [15]. However, on the subjective questions based on different fictional personas, we could not find a consistent model with the highest performance - ChatGPT4o, Clause, and Gemini demonstrated their capability on different fictional personas. Below are our qualitative findings on different commercialized LLM models -

- ChatGPT 4o - providing the customized knowledge base and utilizing the online search option tends to provide better responses for objective questions, but hallucination is still a concern. Keeping the knowledge base and turning off the online search can reduce the hallucinations.
- Claude - there is no option to utilize real-time searches. The answers are more "faithful" to the knowledge base, but the allowed knowledge base size is smaller than ChatGPT, making it challenging to build a domain-specific model on Claude if the knowledge base is larger than the online textual data from an R2 university.
- Gemini - as a model that does not support real-time searches, it is generally good at providing general answers and considering personal needs. However, it does not support structured input (such as Excel files) as part of the knowledge base in the current version.
- Copilot Studio - is currently weak in both correctness and comprehensiveness and is especially weak in answering subjective questions that relate to the needs of a specific persona. However, we see the ambition of the developer - we are hopeful for stronger future versions.

Although this study aimed to create an assessment framework and evaluate the current state-of-the-art LLM models, the pace of innovation in the field will inevitably limit the generalizability of our results and observations. However, our *assessment framework* has a longer impact compared to our findings. Our experiments were done between August and October 2024, and the new OpenAI o1 was not in the scope of this project.

In addition, there will be cases in which the knowledge base cannot be online (either uploaded or shared) due to institutional/cooperation regulations. In this case, the technical paths discussed in this paper will not be applicable. For those researchers/working professionals, the best option will be to create an in-house LLM server.

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